

PhD Fellowship Strategic Basic Research
PROJECT OUTLINE

Changes to previous project proposal

In the current proposal the gaps in the academic literature are described more clearly and more emphasis is put on the scientific contributions of the study and where it fills up the gaps in the literature. In the research methodology, the tools used to develop the algorithmic framework and the specific analyses performed for the validation of the framework are discussed more elaborately. Moreover, in the strategic dimension and application potential section the strategy, concerning the collaboration with the stakeholders, is more concrete. Finally, following the feedback I received from my first application it was decided to submit this proposal to the panel of '*Mechanical Engineering A*' instead of '*Social sciences and humanities*' because it aligns better with the scope.

Rationale and positioning with regard to the state-of-the-art

Over the last decade, the growth of e-commerce has transformed the shopping experience of people entirely. Nowadays, customers expect to conveniently buy (sell) products and services anytime and anywhere through digital technologies and have fast delivery. As a direct result, the competition in the e-commerce sector is fierce, which forces the industry to evolve and expand at a rapid pace, in both developed and developing markets. However, over the last five years, Belgium has lost many distribution centers to neighboring countries, costing us approximately **16.500 in new jobs** creation in 2018. If this trend continues, it is expected that Belgium loses out on around **53.000 new jobs** by 2022 [1]. The existing competitive pressure is driving the urgency for a paradigm shift to more automation and digitalization because of the fear of missing the train [2]. Retailers have to continuously differentiate themselves through quality online services to gain a competitive advantage over their fellow contenders by offering customization of goods, one-day delivery, 30-day return warranty, etc. In order to achieve this, you need a high degree of flexibility in your operations. However, Flemish companies are hamstrung by inflexible Belgian working hour regulations, high wage costs, and a scarcity of blue collar workers in certain regions. Automation and optimization in distribution and manufacturing environments is necessary to provide those services [3].

Flanders is actively trying to attract new e-commerce players to settle in the region. The region is very interesting as a location for logistics activities since 60% of Europe's purchasing power is located in a 500-km radius [4]. However, the inflexible work hours, high wage costs, and labor scarcity discourage the arrival of new business. Hence, it is important for Flanders to leverage the digitalization and automation trends in supply chains. Moreover, given the fact that in many industrialized countries the *workforce is aging, ergonomics and collaborative interactions* between human workers and automated systems should be addressed [5]. Robots can take over dangerous and repetitive jobs of human workers, resulting in a significant cut in costs by replacing expensive people with ever-cheaper machines. The introduction of **automated guided vehicles (AGVs)**, also referred to as *autonomous mobile robots*, for material handling allows facilities to seamlessly sequence different processes between workstations in a fully automated fashion. This automation trend will continue with the emergence of new advanced technologies, e.g., drones, cloud technology, etc. [6].

With modern technologies, robots can perform very accurate movements. However, if such a movement is not the right action at the right time, inefficiencies and wasted movements occur. To avoid this, good control policies for these robots are necessary. The main **control activities** as defined by Le-Ahn and De Koster [8] are as follows: *vehicle dispatching, routing, and scheduling* of AGVs,

deadlock resolution, battery management, and positioning of vehicles. The control decisions all have interdependencies and are visually represented in Figure 1.

This research project will focus on the operational level, more specifically: the vehicle scheduling, dispatching, and routing of **large-scale AGV systems** in **real-world** material handling environments. These interrelated decision problems are crucial for achieving high efficiencies. When material handling robots were first introduced, these problems were rather trivial due to the small number of vehicles operating back in the early 2000s [7]. However, as more large-scale AGV systems are being introduced, the optimization of such problems can no longer be considered as trivial. The vehicle routing and the scheduling respectively determine the route and sequence of loads that a vehicle should handle and the times the vehicle should pick up/drop-off the loads. Originally AGVs used guidance paths incorporated in the infrastructure. However, currently AGVs can navigate freely throughout the facility without being restricted to physical rails or markers. This increased freedom however, comes at the cost of more computationally intensive path finding. In most cases, the dispatching, routing, and scheduling are not as straightforward as just finding the shortest path between origin and destination. In theory, all operations can be completed just as planned with no conflicts. However, many conflicts may arise during operation and these cannot be disregarded. **Commonly known conflicts** are: *vehicle collisions, congestion, livelocks, and deadlocks*. Livelocks and deadlocks occur in situations where vehicles cannot move anymore due to mutual blocking. Therefore, **collision avoidance** and **deadlock resolution** are important considerations during modelling [9].

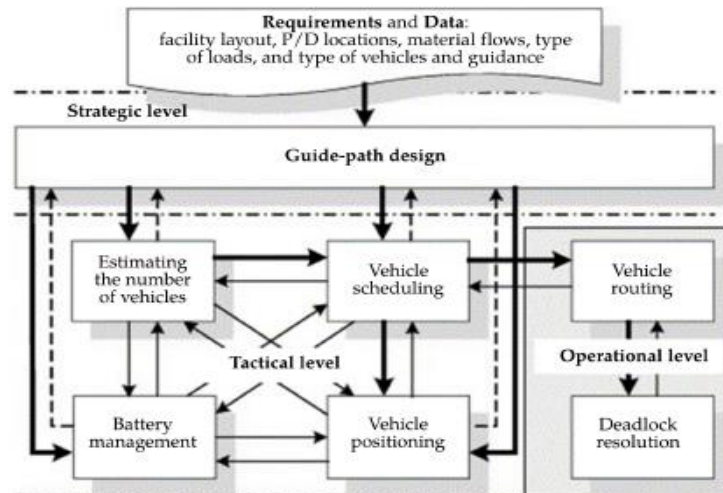


Fig. 1: Decision Framework for design and control of an AGV system. Arrows (thick, thin, and dashed) indicated (strong, less strong, and possible) interactions between the control activities [8].

In literature, two main approaches to solving the AGV control problem can be distinguished: the **static** and **dynamic approach**. Firstly, the static approach in the field of optimization research assumes that certain material handling systems (e.g., AGVs) operate according to a fixed schedule based on off-line data. In many cases, it is possible to find the optimal solution for this problem. Therefore, exact algorithms are highly preferred since the algorithms can guarantee optimality if given enough time. Secondly, with large-scale dynamic problems based on on-line data, exact algorithms typically are not suitable, especially when available computation time is very limited. Therefore, when real-world large-scale problems have to be solved, **(meta-)heuristics** are more appropriate. Compared to exact algorithms they cannot guarantee optimality, but they can reach good solutions in a fraction of the time that an exact method would require [10].

Real-world problems are intrinsically dynamic. When on-line scheduling is concerned, two methods of controlling systems can be distinguished: the *centralized* and *decentralized* approach. A **centralized** architecture has one central control unit that uses global information and typically performs the scheduling, path planning, and coordination between the vehicles [11][12]. The centralized method allows in theory to obtain optimal solutions. However, it has difficulties dealing with large-scale problems efficiently as it takes significant computation time to solve. In contrast, a **decentralized** method does not solve the problem as a whole but splits it up in small sub problems that are assigned to each agent. Various different agents can exist in one system, e.g., the autonomous vehicles, workstations, inventory racks, etc. Every agent decides independently by its own decision logic and makes use of its *local knowledge* to solve the problems it faces, while taking the states of the other agents into account (if known). This enables decentralized systems to scale more easily to larger fleets and require less computational resources. However, such systems regularly result in vehicle deadlocks or collisions because it has no complete view on the system. Some typical examples of decentralized systems are **multi-agent systems (MASs)** and **holonic manufacturing systems (HMS)** [11][13].

The **shortcomings of the state-of-the-art**, which is identified from a systematic literature review, is that the dispatching, scheduling, and routing control activities are often not considered simultaneously. Many researchers focus on one specific control activity, assuming that no interactions occur between them. Nevertheless, in reality they have strong **interdependencies** which affect the overall operational performance [14]-[16]. Hence, in this project, the goal is to integrate these activities allowing for a more realistic solution approach to real-world optimization problems for large-scale AGV systems in distribution and manufacturing environments. To achieve this, I will combine a **centralized scheduling approach** with a **decentralized conflict-free routing approach** into a cooperative framework explicitly taking the **interaction** between the control activities into account. The cooperative behavior between the centralized and decentralized systems is inspired on the idea of reactive scheduling where a baseline schedule is re-optimized during project execution [26]. This concept is well known in the domain of production scheduling, however, to the best of the researcher's knowledge has not been studied yet for the application of AGV scheduling.

Consequently, the **scientific aim** of this research is to investigate the effect of the interdependencies of the control activities on the performance of material handling environments by means of optimization models and experimental validation. The results of this study will allow to provide guidance on how to schedule and control resources like autonomous vehicles such that not only the system they operate in runs more cost effective but also more time efficient. The optimization of operational efficiency is highly important for e-commerce businesses to remain competitive.

Scientific research objective(s)

With the outcome of this project I will be able to answer the research question "**Can a hybrid control framework outperform both the centralized and decentralized models for large-scale dynamic problems?**". To achieve this goal, I will develop a modular control framework for large-scale AGV systems solved in real-time. The resulting framework has to be able to deal with the control activities in reasonably short computation times. Furthermore, I will validate the framework in computational experiments using instances generated from real-world cases. The project's objectives are:

Objective 1: The generation of realistic benchmark instances

In literature, good benchmark instances for real-life problems are limited. The available instances are **very small in problem size** or they do not represent actual real-world cases well [17][18]. Furthermore, these instances were created as static scheduling problems where all information is known

beforehand. In such static cases, approaches designed for dynamic environments cannot show all of their capabilities, and therefore, dynamic approaches are dominated in academic literature by exact solvers [19]. **Dynamic** cases occur in practice, for instance, whenever the demand for specific goods stored in a distribution center fluctuates heavily and arrives at unknown points in time. Another example to illustrate the dynamic nature of real-world situations is the occurrences of machine breakdowns or uncertain truck arrival and departure times directly affecting order cutoff times. Hence, the first research objective of this project is to **generate benchmark instances** for realistic large-scale dynamic AGV problems.

Objective 2: Development of a modular control framework

The second research objective of this project is the development of an effective modular control framework for large-scale autonomous mobile robots in distribution or manufacturing environments. As mentioned in the previous section, the framework will consist of a cooperative centralized scheduler and a decentralized dispatcher with conflict-free routing considerations. The framework has to be modular, so that it can be flexibly altered depending on the specific problem characteristics. Several considerations are taken into account, e.g., collisions, deadlocks, congestion, job consolidation, etc. Incorporating these considerations into the framework is key since decentralized problems typically struggle with the avoidance and resolution of collisions and deadlocks. **The main contribution of this research project is to simultaneously solve the scheduling, collision-free routing, and dispatching activities for large-scale dynamic problem instances.** As mentioned previously in the state-of-the-art, most academic papers consider only one or two activities simultaneously, but almost never deal with all of them at the same time.

Objective 3: Incorporation of speed optimization

In large-scale material handling systems, especially when a decentralized control approach is concerned, collision avoidance and deadlock resolution are critical considerations. Implementing speed optimization allows for the vehicles to retain their velocity at conflict points, where normally the vehicle has to stop to avoid a collision. A secondary advantage is that it can reduce the overall energy consumption, for which speed is known to be a determining factor [21]. Furthermore, my systematic review has shown that many academic papers make the assumption that vehicle velocity is a constant and deceleration and acceleration are considered as negligible. Ignoring these effects hinders the applicability of the algorithm solutions in real-life situations. Therefore, I will implement a **speed optimization strategy** in the proposed framework.

Objective 4: Incorporation of heterogeneous, multi-load AGV capabilities

Considering the constant technological evolution of AGVs, various types of AGVs exist and can be distinguished based on their functionality, typical functionalities are: towing AGV, pallet trucks, fork truck, unit load AGV, and assembly line AGV [20]. Some are capable of transporting different amounts of loads at a time. Whether an AGV should be able to handle single-loads or multiple-loads is a decision that has to be made in advance [14]. The state-of-the-art shows that many papers do not consider fleet heterogeneity or multi-load capabilities of AGVs, although they can considerably increase the efficiency and throughput overall [23]. The reason why researchers generally tend to avoid this may be because the implementation of these factors greatly increases the complexity of the problem. Thus, as my fourth objective I will incorporate **fleet heterogeneity and multi-load carrying capabilities** into the proposed framework.

Research methodology and work plan

Every good research starts with an in depth review of existing literature in order to get a good grasp on the state-of-the-art of the specific research domain. Hence, I made a **systematic review** based on the research protocol proposed by Wright et al. [22]. It allows for the reduction of the population size to a final sample size of only adequate literature for the review by applying *inclusion* and *exclusion* criterion. A systematic review is necessary to identify **research gaps** and justify further research. The results of the review were used to position this proposal in the state-of-the-art and formulate future research objectives. To achieve the previously formulated objectives, each research objective was assigned to a work package, except for the second objective. The development of the modular framework was split over three work packages. Since no standard tool kit exists for metaheuristic development, I will implement my metaheuristic and agent based systems in **Python**. The effectivity of the algorithms will, for small instances, be compared to exact solutions found by exact solvers such as *Cplex* or *Gurobi*. Figure 2 provides a schematic overview of the work packages and their interactions.

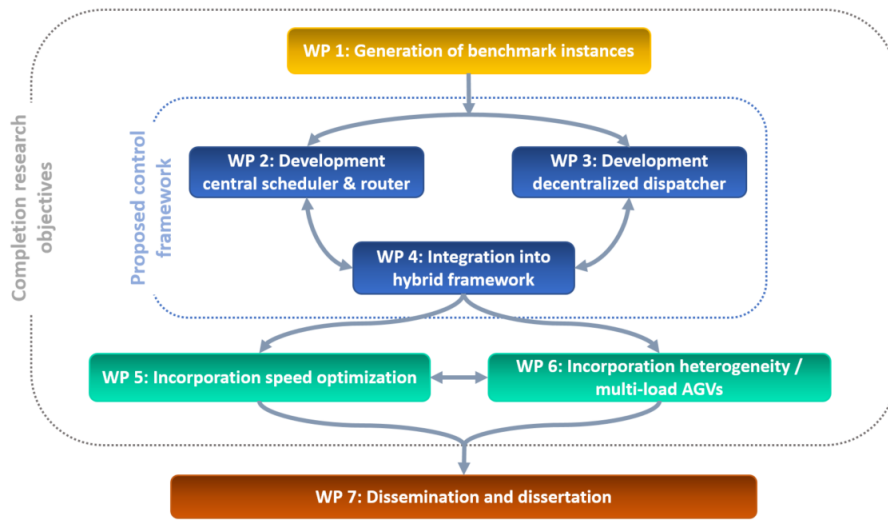


Fig. 2: Project flowchart

WP 1: The generation of realistic benchmark instances

As mentioned above, in the academic literature the presence of realistic benchmark instances is limited. Typically, either the problem sizes are too small and/or many simplifying assumptions are made so they do not represent reality well. The *first step* of this work package will consist of gathering those benchmark instances from the literature that best represent reality with regards to complexity and/or scale. The *second step* involves gathering data from actual distribution or manufacturing environments and generating realistic instances from this data. As for the *last step*, not only actual real-life instances are interesting, but also some **edge-cases** will be generated. The aim of these cases will be to test the framework to more extreme situations and analyze its performance.

The generated instances will concern various types of warehouse and manufacturing systems. For instance, efficient warehousing can be achieved through different approaches depending on what works best for the type of warehouse and the type of goods that are being handled. A *picker-to-parts* system is one possible warehousing system where a human picker has to physically move to the locations of the pick orders located in the storage area. This can also happen in collaboration with AGVs. Conversely, a *parts-to-picker* system is more automated where the pickers, typically human, are designated to stations located around the storage area and the autonomous robots (AGVs) can roam around in the storage area to retrieve the requested goods [3]. Here, the walking distance of the

human pickers is greatly reduced and the productivity increased overall. The same approach in generating benchmark instances will be taken for manufacturing systems. By the end of this work package, several different benchmark instances will be created which can be used to validate the modular framework. Additionally, and not unimportantly, it will make it easier for the **research community** to focus their efforts on these realistic and large-scale instances.

The research group where my research takes place, **Centre for Industrial Management / Traffic and Infrastructure (CIB)**, has already established good collaborative relationships with several companies (e.g., Logflow, DPD, Miebach, Easy Systems, etc.) active in distribution logistics and manufacturing. These companies will be able to evaluate the results and provide valuable feedback from practice. Moreover, in my supervisory committee a professional from one of these companies (Miebach) is seated and the discussion on sharing data and insights is already ongoing.

WP 2: Development of a centralized scheduler and router for large-scale dynamic problems

In work package 2, the development of a centralized control model is the main focus. A central controlling model has the task of finding near-optimal solutions, the state-of-the-art has shown that exact algorithms are currently inappropriate for large-scale problems. Hence, heuristics and metaheuristics will be used. The model will have to account for the control activities simultaneously. For the simplest problem types, for instance a straightforward parts-to-picker system, such a central scheduler needs to deal with **(1)** assigning which orders are going to be handled by which pickers, **(2)** assigning the retrieval of all the orders' order lines to the AGVs, **(3)** sequencing the retrieval jobs for each AGV, **(4)** sequencing the pick jobs for each picker, and **(5)** determining a detailed routing for the AGV fleet. These 5 problems are highly interdependent and not easy from a computational point of view (e.g. the AGV assignment & sequencing can be considered as a Vehicle Routing Problem, a notorious NP-hard problem). In order to validate the centralized model the resulting solutions, typically these are vehicle and order sequences, are simulated in order to determine the systems' performance. Several possible **metrics** can be calculated that together express the performance, e.g., *robot utilization rate, workstation utilization rate, average order cycle time, makespan, total distance travelled, etc.*

Special focus will be given to designing this framework in modules. In this way, the solution framework can be easily modified to be applicable to various types of distribution and manufacturing environments. In some cases different algorithms may be more suitable than others to solve specific problems since heuristics are problem dependent. The modularity of the model allows to insert and remove certain parts of the coding to ensure an effective and efficient solution approach. The meta-heuristic(s) that will be used in the central scheduler make use of many parameters that require **parameter calibration** to ensure good quality solutions. This is best performed using a systematic approach [27]. However, this requires a large number of experiments for which the **Flemish Supercomputer Center** is well suited. The *high performance cluster* (HPC) allows to run the algorithmic framework on multiple experiments in parallel which I have already performed for similar metaheuristics.

WP 3: Development of a decentralized dispatcher

In this work package, the development of a decentralized dispatcher with collision-free routing consideration is studied. The goal here is the same as in work package 2, where all the control activities have to be handled simultaneously by the model. To achieve this, I intend to create a multi-agent based model, consisting of autonomous agents that coordinate and interact with one another and the environment. The agent-based architecture is a more advanced solution approach compared to

classical dispatching rules. One of the key features of multi-agents are the negotiation mechanisms that allow the agents to share information [28]. **Auction based** methods, which are negotiation strategies, will be considered in this study. This method is inspired on the principles of auctioning, when a task is put for auction, the vehicle agents can place a bid based on their current situation, e.g., the distance from their current location to the task destination. The auctioneer, e.g., a workstation, gathers these bids and decides which agent is best suitable for the task. This process continues until all tasks are completed. Whenever an unexpected event occurs (e.g. machine breakdown), re-auctioning can be initiated in order to adapt to the new situation. The validation of the model will be done using the same metrics as mentioned in work package 2 to ensure a fair comparison.

WP 4: Integration of both centralized and decentralized approaches

“No plan of operations extends with any certainty beyond the first contact with the enemy” (Helmuth von Moltke). In warehousing/manufacturing operations, unexpected events and statistical fluctuations on processing and travel times can be considered as the enemies of our planning. This implies that any schedule determined by a scheduler, be it central or decentral, will be **out of date as soon as it is released** to the warehouse or production floor. Furthermore, it is crucial to recognize that the envisioned scale will prohibit asking the central controller to re-optimize the complete schedule as soon as deviations are communicated. Here, we acknowledge the strength of decentral systems and assume that short-term decisions can be taken locally as long as they do not impact the overall schedule too much.

For the fourth work package, I will integrate the previously developed models into one hybrid control framework, in this study hybrid refers to the combination of a centralized and decentralized solution approach. This work package will be a very iterative process in order to figure out which algorithms complement each other the best. Particular focus will be given to the feedback loop between the two models. The communication through the feedback loop handles the cooperative aspect of the framework between the two models. Here it is important to determine which information has to be communicated in order to guarantee a good alignment between both models.

One envisioned way to combine the models is by first letting the central model generate a schedule and set of routes. This is launched in the simulation environment where the decentral agents try to follow the near-optimal schedule. However, when a deviation (e.g. machine breakdown) occurs, the agents negotiate amongst each other to make alterations to the existing schedule. After some time, these deviations on the schedule will have accumulated making the original schedule outdated. The central model is then triggered to generate a new schedule and set of routes. For this to work, a frozen time period may be required where the decentral model completely takes over the scheduling, routing, and dispatching. During this time the central model generates the new schedule and route with the tasks left after that frozen period. Numerous experiments will be required to ensure that the dispatcher follows the near-optimal schedule well. I envision it to be the hardest work package with the highest risk and it will likely require the most time to complete. The **validation** of the hybrid framework will happen using the same metrics as in previous work packages.

WP 5: Incorporating speed optimization

Traditionally, models for routing and scheduling make the assumption that vehicle velocity is constant. However, in reality the velocity is a variable that can be altered depending on the situation, it can be introduced as a decision variable. Speed optimization can bring many benefits concerning *conflict avoidance, retrieval execution times, and energy consumption* (battery management). By varying the velocity of the vehicles, it becomes possible to control it in such a way that it no longer has to come

to a complete stop at potential conflict areas but retain some of its initial velocity. This requires considering the acceleration and deceleration profiles of the vehicles (in loaded and unloaded state).

The incorporation of speed optimization into the model can result in a more effective and faster way to resolve and avoid deadlocks because unnecessary waiting at nodes can be avoided [21]. This is interesting because decentralized approaches generally are more prone to deadlocks. However, determining the optimal schedule will be somewhat difficult because it is highly dependent on the speed of the AGVs. Therefore, the optimal schedule has to be based on the optimal speed of every AGV utilized in the environment. The **Speed Optimization Problem (SOP)** can be referred to as a sub problem, which is to determine the optimal speed along a route of an AGV [24]. I will research (meta-)heuristic or approximation algorithms to solve this sub problem. Some heuristics dealing with the SOP already exist and can serve as a source of inspiration. I expect that including speed optimization explicitly in the centralized scheduler would be too computationally expensive. An approximation approach might possibly alleviate some of this computational burden. In the hybrid approach, I expect that explicit modelling of the speed optimization problem is feasible on the agent based level.

WP 6: Incorporating heterogeneity, multi-load AGVs

In work package 6, I will incorporate heterogeneous and multi-load capabilities of vehicles into the hybrid model. Here it is important to develop appropriate dispatching rules for the selection of available vehicles to ensure the effective control of multi-load AGV systems. The literature review has shown that the selection of dispatching rules influences the overall throughput and lowers the makespan of the performed jobs. Moreover, it can significantly reduce the complexity in routing and scheduling of multi-load AGVs [25].

Taking the example of the parts-to-picker system, where only single loads are concerned, the retrieval of orders can be seen as a **multi-Travelling Sales Person (TSP)** with an asymmetric distance matrix. An AGV with single load carrying capacity can only accomplish one task at a time before having to return back to the depot, here a depot is a predetermined location where the AGV has to deliver its request. Alternatively, vehicles with multi-load capabilities can pick up additional loads while on its route to accomplish a previously assigned task. The introduction of the multi-load AGVs allows for a significant reduction in the number of moves that a multi-load AGV has to perform to accomplish the same amount of tasks as a single-unit carrier. However, the scheduling, routing, and dispatching become more complex since the sequencing of the tasks within one retrieval route and the choice of which vehicle has to perform a task become more critical.

WP 7: Dissemination, dissertation writing, and completion of doctoral program

Finally, when all the six work packages are concluded, the dissertation writing can start. To ensure its successful completion, I will follow additional courses provided by the KU Leuven to enhance my scientific writing and other transferable skills. Some examples for relevant specific courses are: *Scientific Integrity, Writing the PhD Thesis, Intensive Academic Writing Course, etc.* The completion of this final work package will result in the conclusion of the research project.

Project management

The Gantt chart (Figure 3) visualizes the expected progression of this 4-year research project in more detail. At the beginning of every work package a literature review about topic at hand will be made. Hence, the literature survey is not included as a separate work package, but rather as an ongoing process. As mentioned, before during the fourth work package a iterative process happens over work package 2, 3 and 4. After the completion of a work package some time is planned to report my findings

up until that point and to write a research paper for a journal or conference (e.g. *ORBEL*, *EURO*, *ICOR*, ...). Finally, to complete this last work package at most five months are scheduled, time for courses on relevant transferable skills and dissertation writing skills are included within this time period.

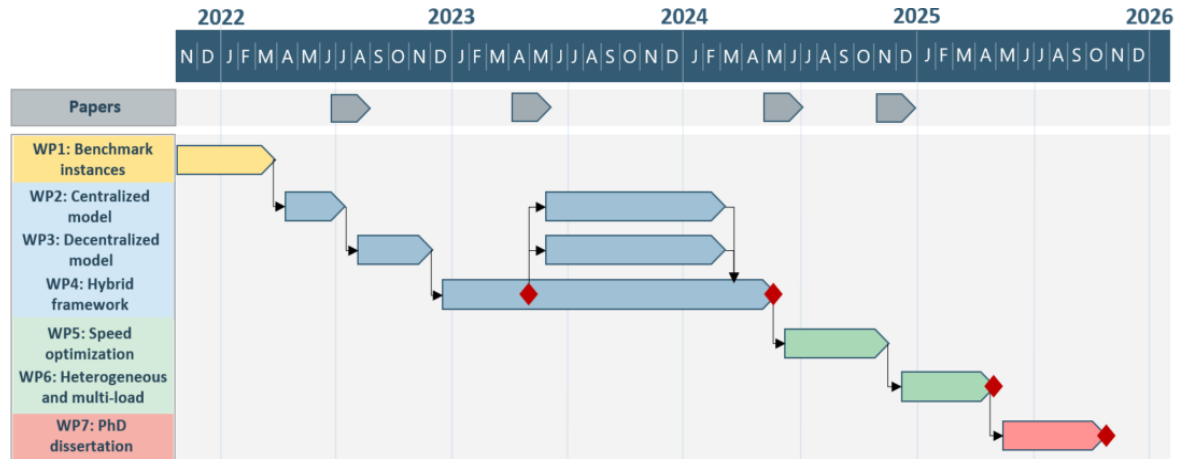


Fig. 3: Gantt chart showing the planning of the work packages and their interactions. The diamond indicates a milestone.

Risk management

The feasibility of this research depends not only the skills and expertise of the researcher and his entourage, but also on the potential external risk factors. Therefore, the development of mitigation strategies is important. To ensure successful progression of the project the researcher will be guided by a professional supervisory committee, consisting of a promotor (Reginald Dewil, expertise in scheduling and routing algorithms), one other academic from the KU Leuven (Tom Holvoet, expertise in computer science), and a professional (Marc Meuleman, managing director at Miebach Consulting Benelux). Moreover, the risks are partially mitigated by good cooperation with other research groups, for instance the MECO research group that work on similar or relevant problems. I have identified several risk factors for this research project, which are discussed in more detail below.

1. The first risk that has to be considered is the occurrence of delays or incomplete deliverables of the work packages. Some packages can only start when its predecessor finishes. For instance, if either work package 2 or 3 are delayed, this also affects the finishing time of work package 4. The risk can be minimized by strictly monitoring the progression of this process, by doing so the project can be kept on track. Work package 2 and 3 can be worked on in parallel. Hence, if one package experiences difficulties, the other one can start without losing too much time. Moreover, some models exist that deal with specific problems relevant for this project, even though simplifying assumptions are typically made. These can be used as an inspiration source for the proposed control framework. The most critical risk factor for this research is the integration of both models into one cooperative framework. To minimize the risk of failure, the most time is foreseen to complete it.
2. Another possible risk concerns the participation of stakeholders. For work package 1 data has to be gathered in order to generate instances that represent the real-life conditions accurately. The stakeholders in this case are the players in the logistics industry or manufacturers. However, if they are unwilling to share their data to build these instances or simply no real-life data is available, this can affect the progression of the work package significantly. However, for this project there is limited risk that it will seriously affect the

outcome since the control framework could also be validated on self-generated instances that closely resemble real-life scenario's, even though this would be less ideal.

3. A third risk is that the generated benchmark instances have poor quality due to incomplete data (from the stakeholders). If the data insufficiently covers all the aspects of the problem this may lead to an inaccurate representation of the real-life problem. This can occur whenever several simplifying assumptions have been made or data that has been withheld. A way to mitigate this risk is to further extend the datasets with other real-life data to approach the reality as much as possible.

Strategic dimension and application potential

With the rapidly growing e-commerce business, even more so due to the COVID-19 pandemic, the competitive pressure among these businesses becomes fiercer. Although Belgium is a small country, its location makes it extremely attractive for these types of distribution companies. However, high wage costs and blue collar labor scarcity makes these companies choose for the neighboring countries, e.g., France, Germany, and Netherlands. Flanders acknowledges that there is still a significant growing potential and is actively trying to attract new e-commerce business to settle in the region [4]. To prevent this trend, the paradigm shift to more automation and digitalization has to be promoted. Overall operational costs of material handling can be significantly reduced as a result of more efficiency and flexibility since mobile robots are able to work 24/7. These main strong points can level the playing field with our neighboring countries with respect to established distribution centers or manufacturing centers [1]. Some might argue that replacing human workers with autonomous robots leads to less jobs. However, because e-commerce businesses tend to settle in our neighboring countries, these jobs are lost in Flanders anyway. Moreover, **low quality** (*mundane and repetitive*) jobs can be replaced by **higher quality** jobs (e.g. *programming, maintenance, etc.*), which in turn can increase employee satisfaction and retention. On top of that, in order for warehouses and manufacturers to operate well, various other businesses are involved, not in the least transportation, ..., this promotes job creation. Overall more automation brings opportunities for job creation, and therefore, it brings **economic added value**.

The model that will be developed has to work for real-world applications and is not only relevant for scientific purposes. By considering large-scale dynamic environments with constraints inspired by practice in real-time, there is a direct application potential. Moreover, the proposed project aligns with the fourth strategic VRWI cluster (*Urban Planning, Mobility Dynamics, and Logistics*), formulated for the priorities in science, technology, and innovation towards 2025 in Flanders. More specifically, the '*Synchromodality*' and '*Intelligent Transport Systems*' priorities correspond with the content of the proposed topic. The handling of goods happens through **synchromodal transport** (e.g. airplanes, freight carriers, trains, etc.) which has to be properly synchronized with one another. For instance, the synchronization of distribution centers is critical and can be achieved with automation and effective control algorithms. The implementation of **Intelligent Transport Systems (ITS)** in the proposed control framework is key in ensuring the efficient functioning of the material handling system.

The **main target group** of this project are the consultancy companies like Miebach and Logflow who specialize in logistics. The reason why I do not intend to directly approach potential end users, like for example Amazon, is because they tend to be very protective about how their systems run. Moreover, generally they are not eager to share data with a doctoral researcher. However, the previously mentioned consultancy companies do have very good connections with these e-commerce businesses and can provide (anonymized) data and cases. Presenting the results and insights from the proposed framework to them first has a higher potential of eventually reaching the actual end users. Currently,

I am in discussion with Miebach, this contact also belongs to my supervisory committee, concerning the data sharing. For instance, about what specific data (*e.g. order arrival time, number of operational AGVs, number of inventory pods, employee operation regimes, etc.*) I require to generate benchmark instances. Moreover, my promotor and I are also in contact with Logflow. Through the discussions with these companies I aim to get a clear image of what the needs are of all the stakeholders. These needs can directly be taken into account by incorporating them into the objective function of the model, by doing so the schedules and routing are optimized to that objective. Based on the experience of my research group (CIB), I aim to develop a framework that can reduce the *operational* and *capital* costs by **10 to 15 %** compared to commercial systems.

This research is a perfect opportunity to broaden the knowledge base of the research operation community, because the potential of combining centralized and decentralized approaches is understudied. Finally, an important aspect of the proposed hybrid control framework is that it will be built in a modular fashion. Hence, this work is not only applicable for warehouse or manufacturing environments, but after some adaptations to the model, it can also be used to solve optimization problems in container terminals, cross-docks, and other environments. Therefore, this research has a **wide application potential** and many follow-up projects (*e.g. theses, PhD projects, and post-doctoral projects*) can be created. As a post-doctoral follow-up project I aim for a **VLAIO innovation mandate**, where the framework can be developed to a *robust commercial software (economic valorisation)*.

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